

EDA Interpretation

Top 15 scorers

High PPG does not equal high efficiency. SGA and Jokić score fewer points than Luka on average but shoot far more efficiently. Jaylen Brown is the least efficient of the top scorers despite high volume. Even amongst the most reputable scorers who face tough defensive coverages and take harder shots than the average player, volume and efficiency alone are insufficient to determine true scoring value. This emphasizes the importance of this model, as by incorporating shot difficulty and situational context, we can build a more complete picture of how good a scorer each player truly is.

Efficiency vs Usage Rate

As usage rate increases, players tend to score more points, reflected by the brighter colors on the right side of the chart. This is consistent with higher opportunity leading to more shot attempts and higher PPG. Variance is highest among low usage players on the left side of the chart. This makes sense as low volume players have smaller shot samples, making their efficiency numbers less reliable. As usage increases, efficiency regresses toward the league mean and higher quality players remain efficient at high volume, but not astronomically above average due to the increased difficulty of shots and greater defensive attention they face.

Severe outliers in true shooting percentage, both above and below the mean, are concentrated among low usage players. High efficiency outliers at low usage likely benefit from offensive gravity created by teammates, and low efficiency outliers at low usage are likely poor scorers who rarely shoot for good reason. Neither extreme is reliable for evaluating true scoring ability, which further supports the need to adjust for usage when building a scoring impact model.

Scoring Distribution

The distribution of points per game is right skewed, with a heavy tail on the right side. Most NBA players hover around 5-15 PPG, making 20+ PPG scorers and especially 25-33 PPG scorers statistical outliers. This makes high volume scoring that much more impressive and valuable.

Modern NBA offenses emphasize depth and ball movement, while defensive schemes have become increasingly versatile and complex. Despite this, a select few elite scorers consistently average high PPG, which speaks to the difficulty and rarity of what they do. A scoring model must account for this skew as the difference between a player going from 8 to 13 PPG is common, while the jump from 20 to 25 PPG represents a much rarer and more meaningful level of offensive ability.

Correlation Heatmap

Scoring volume and efficiency have a weak, but positive correlation, as only 6.25% of the variation in efficiency is explained by the scoring volume. Overall, the weak positive correlation is relatively consistent with confounding by player talent, as more talented scorers tend to have higher usage and better efficiency, explaining the slight positive relationship. However, the relationship being weak confirms that volume alone does not explain efficiency. Shot type, difficulty, and defensive context account for the remaining variation, hence is exactly what this model aims to capture.

FGA and PTS have an extremely strong correlation ($r = 0.98$), confirming that scoring is primarily a volume stat, more shot attempts directly translates to more points. This highlights the need to adjust for opportunity when

evaluating scoring quality.

TOV and USG_PCT have a strong correlation ($r = 0.80$), which is expected as high usage players handle the ball more and therefore turn it over more. This is not necessarily a criticism on those NBA players, it is a byproduct of opportunity.

OFF_RATING has relatively weak correlations with most individual stats, suggesting that offensive rating captures team level context that individual box score stats do not fully explain. This makes it a useful independent feature in the model.

TS_PCT and USG_PCT have virtually zero correlation ($r = 0.0$), confirming that efficiency and usage have no linear relationship at the league level. Elite players who maintain high efficiency at high usage are genuinely beating the expected tradeoff.